Figure 1

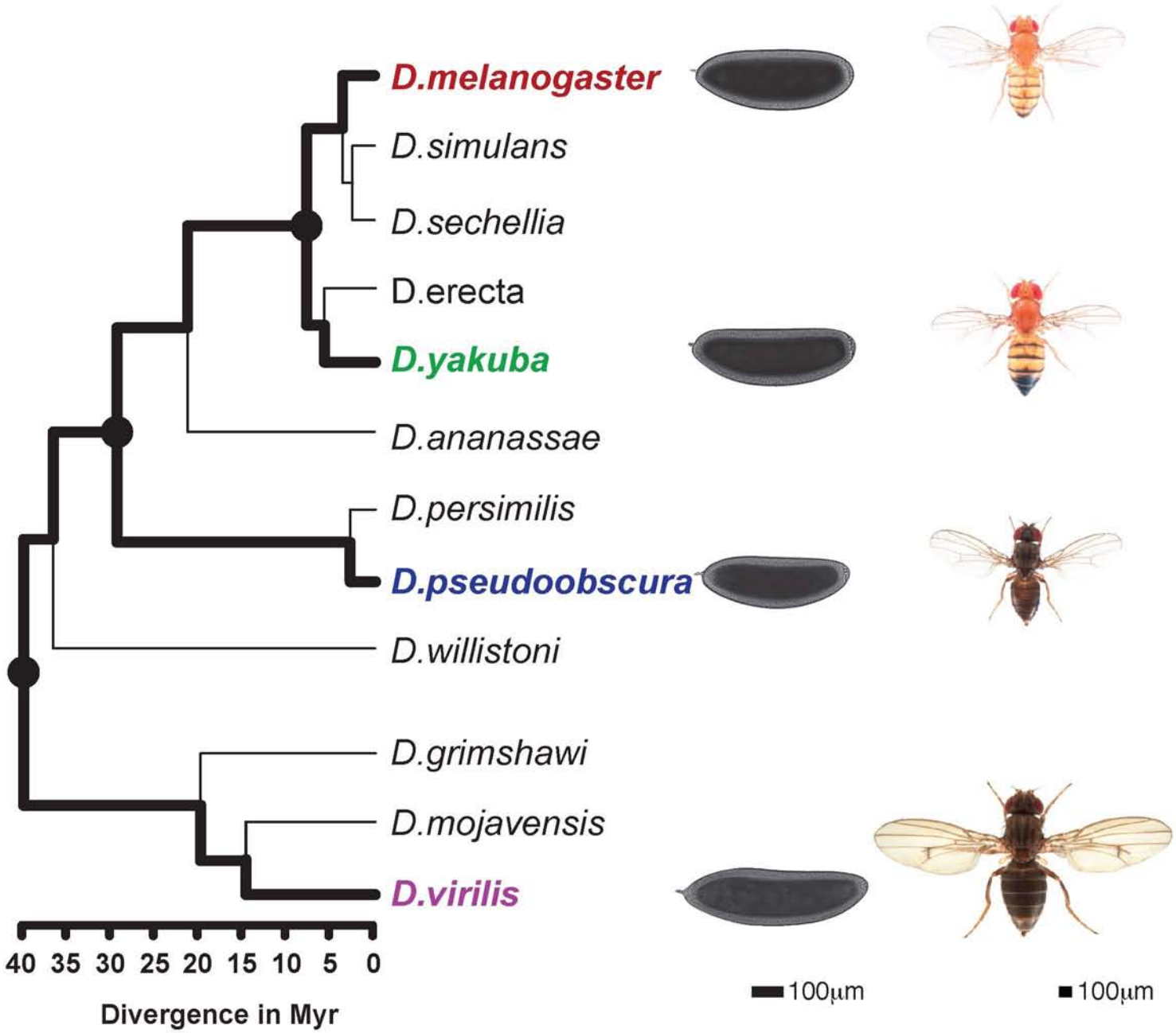

D.melanogaster
D.simulans
D.sechellia
D.erecta
D.yakuba
D.ananassae
D.persimilis
D.pseudoobscura
D.willistoni
D.grimshawi
D.mojavensis
D.virilis
40 35 30 25 20 15 10 5 0
Divergence in Myr
100μm
100μm

Figure 2

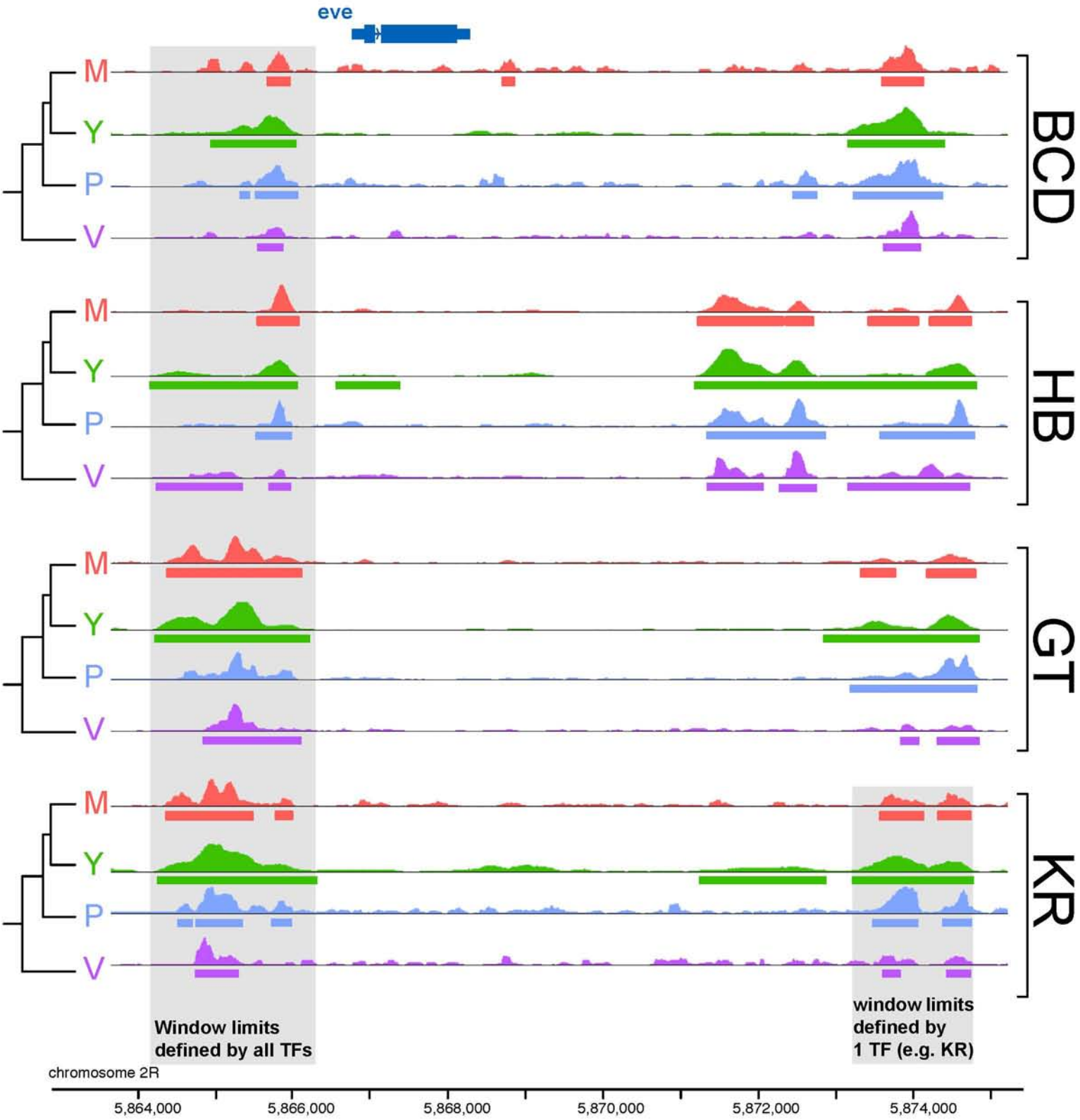

eve
M
Y
P
V
BCD
HB
GT
KR
Window limits
defined by all TFs
window limits
defined by
1 TF (e.g. KR)
chromosome 2R
5,864,000
5,866,000
5,868,000
5,870,000
5,872,000
5,874,000

Figure 3

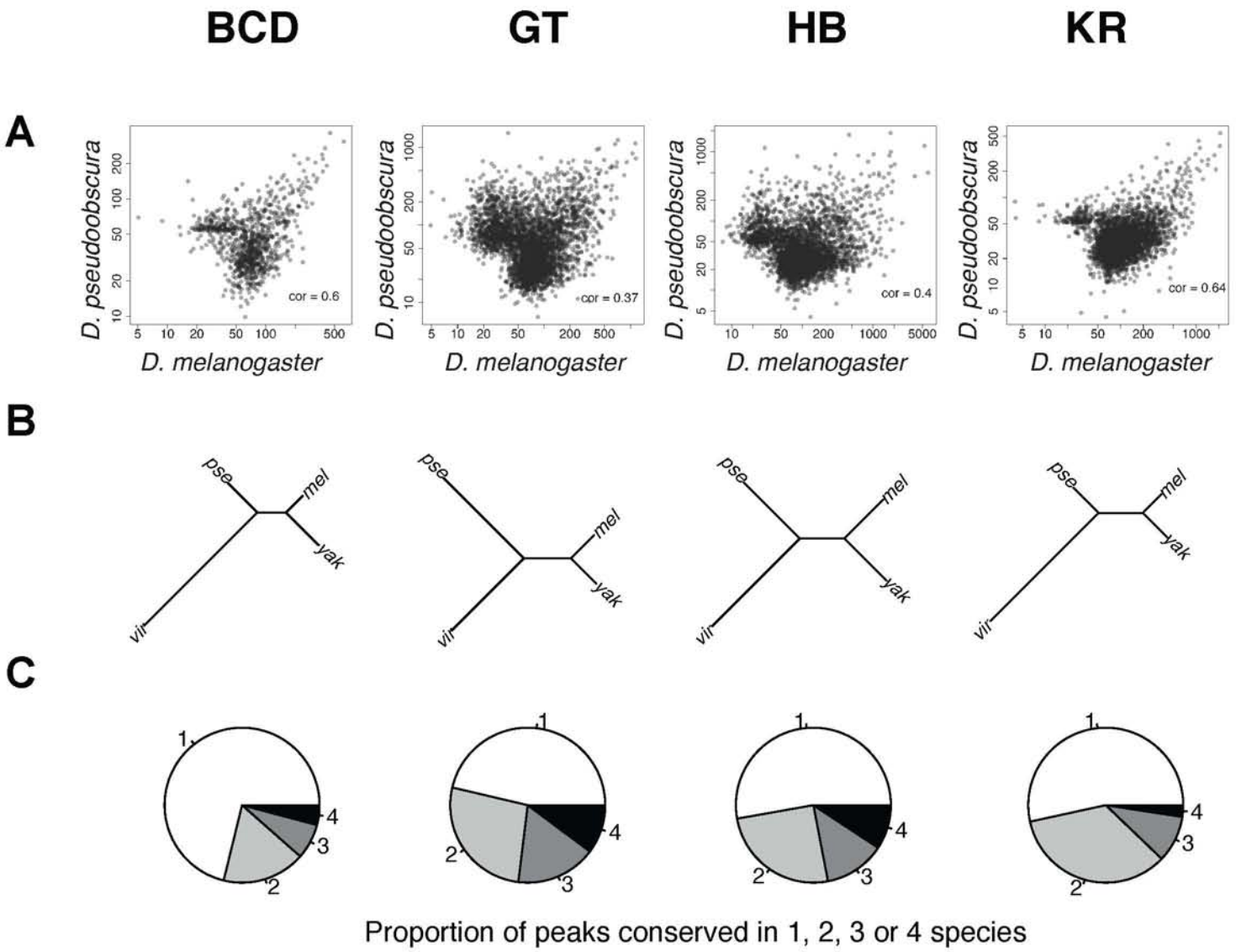

BCD
GT
HB
KR
A
D. pseudoobscura
D. melanogaster
cor = 0.6
cor = 0.37
cor = 0.4
cor = 0.64
B
pse
mel
yak
vir
C
1
2
3
4
Proportion of peaks conserved in 1, 2, 3 or 4 species

Figure 4

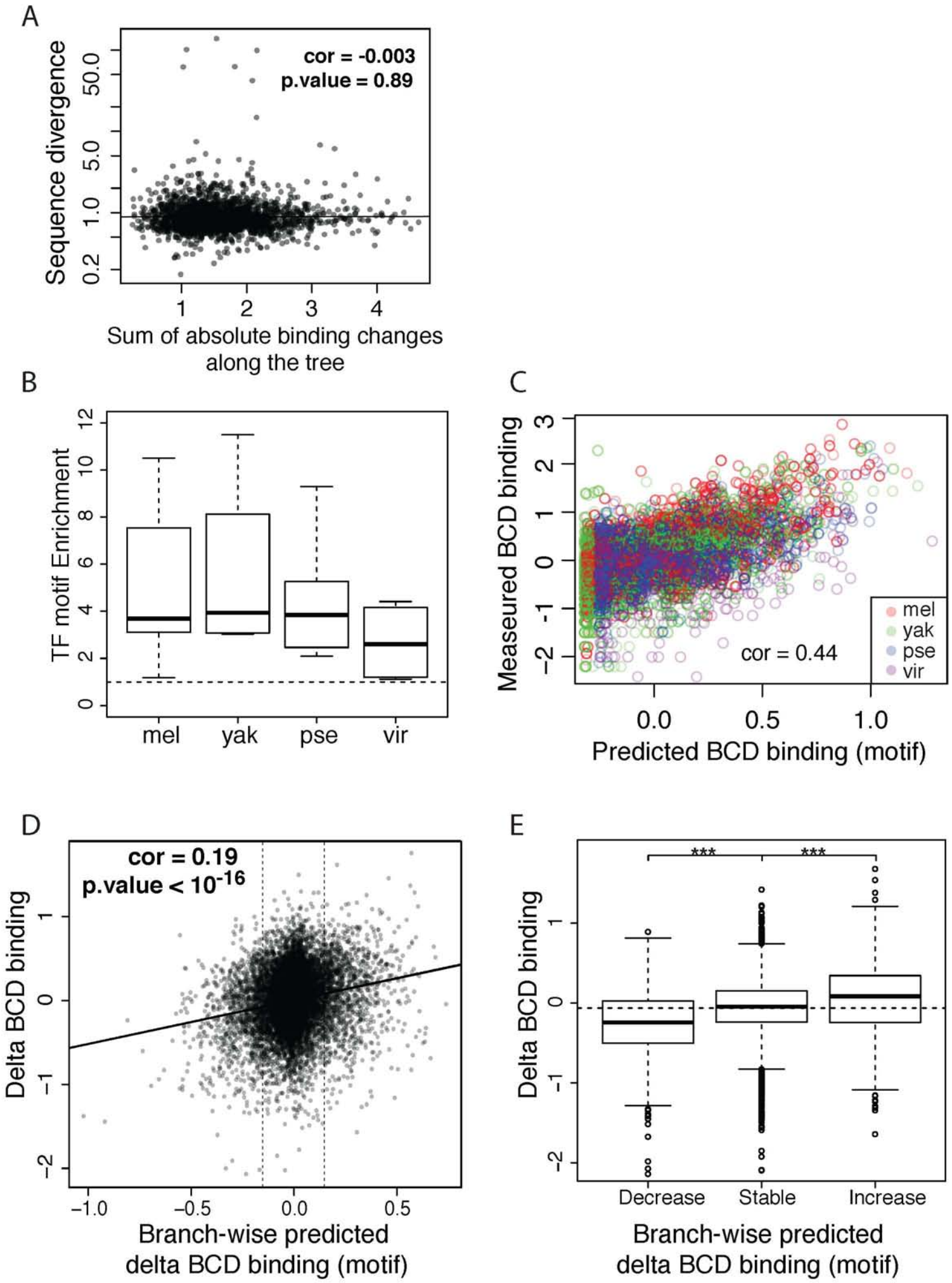

A
cor = -0.003
p.value = 0.89
Sequence divergence
0.2
1.0
5.0
50.0
1
2
3
4
Sum of absolute binding changes along the tree
B
TF motif Enrichment
0
2
4
6
8
10
12
mel
yak
pse
vir
C
Measeured BCD binding
−2
−1
0
1
2
3
cor = 0.44
mel
yak
pse
vir
0.0
0.5
1.0
Predicted BCD binding (motif)
D
cor = 0.19
p.value < 10^-16
Delta BCD binding
−2
−1
0
1
−1.0
−0.5
0.0
0.5
Branch-wise predicted delta BCD binding (motif)
E
***
***
Delta BCD binding
−2
−1
0
1
Decrease
Stable
Increase
Branch-wise predicted delta BCD binding (motif)

Figure 5

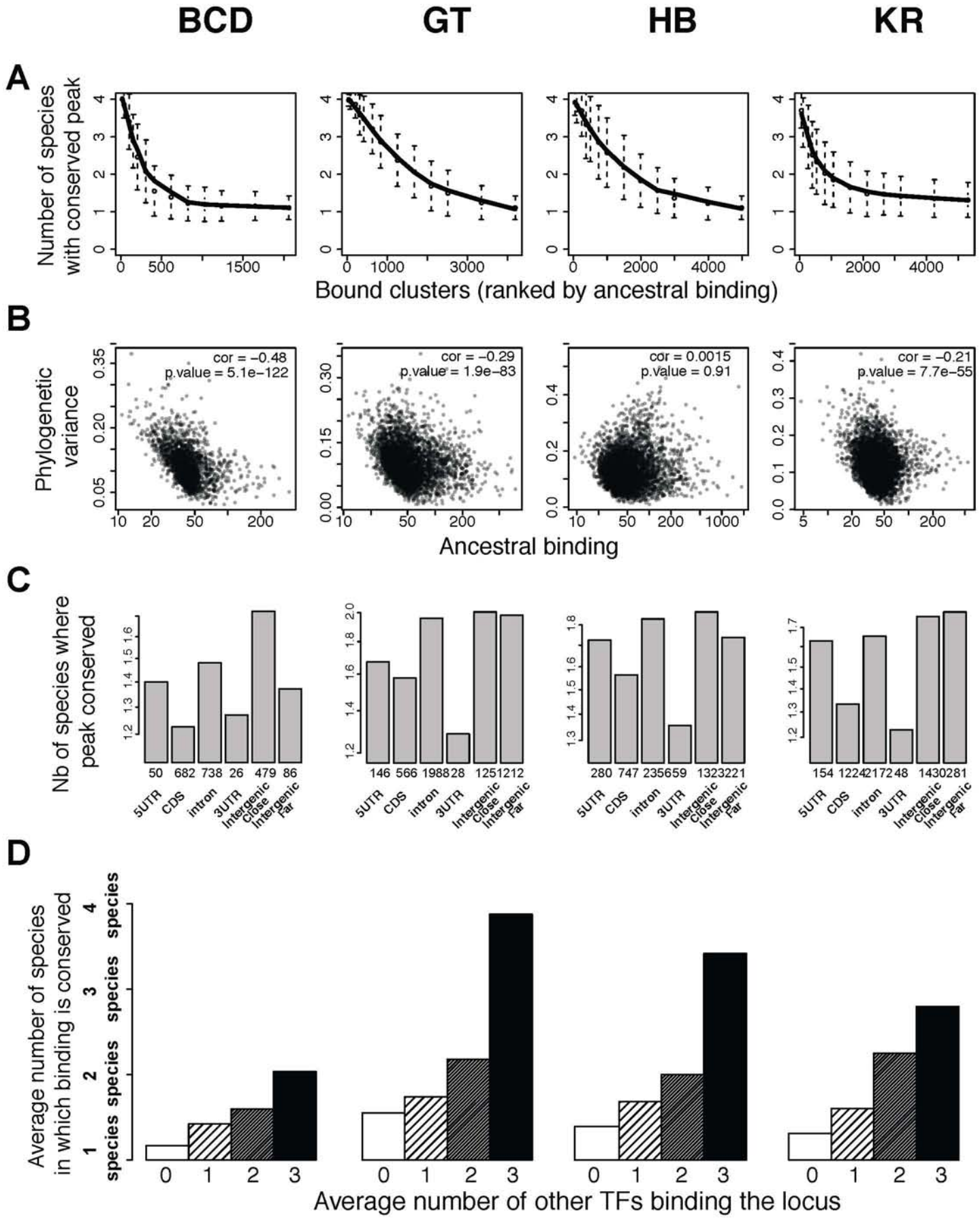

BCD
GT
HB
KR
A
Number of species with conserved peak
Bound clusters (ranked by ancestral binding)
B
Phylogenetic variance
cor = −0.48
p.value = 5.1e−122
cor = −0.29
p.value = 1.9e−83
cor = 0.0015
p.value = 0.91
cor = −0.21
p.value = 7.7e−55
Ancestral binding
C
Nb of species where peak conserved
50 682 738 26 479 86
146 566 1988 28 1251 212
280 747 2356 59 1323 221
154 1224 2172 48 1430 281
5UTR
CDS
intron
3UTR
Intergenic Close
Intergenic Far
D
Average number of species in which binding is conserved
1 species
2 species
3 species
4 species
Average number of other TFs binding the locus

Figure 6

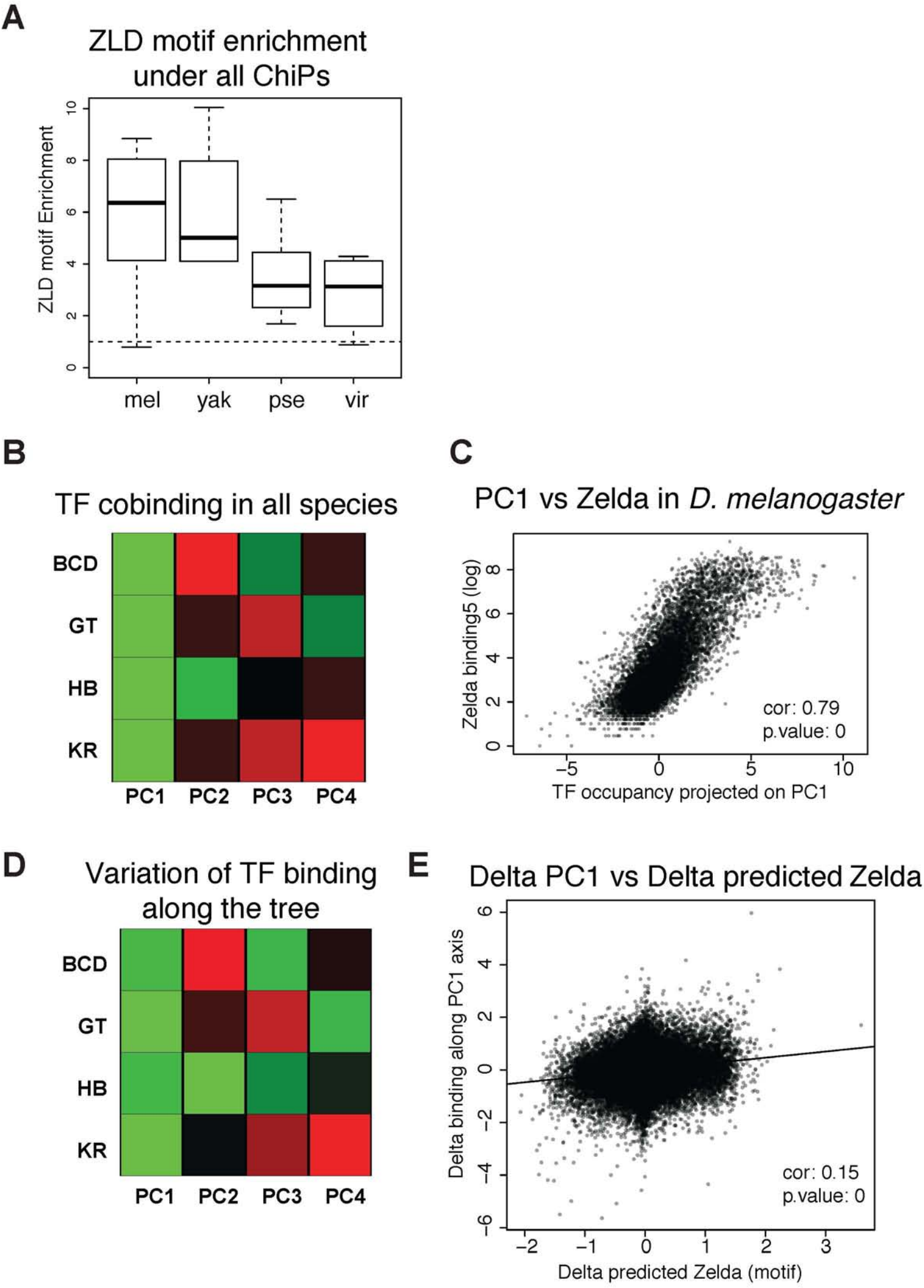

A
ZLD motif enrichment under all ChiPs
ZLD motif Enrichment
0
2
4
6
8
10
mel
yak
pse
vir
B
TF cobinding in all species
BCD
GT
HB
KR
PC1
PC2
PC3
PC4
C
PC1 vs Zelda in D. melanogaster
Zelda binding5 (log)
0
2
4
6
8
−5
0
5
10
TF occupancy projected on PC1
cor: 0.79
p.value: 0
D
Variation of TF binding along the tree
BCD
GT
HB
KR
PC1
PC2
PC3
PC4
E
Delta PC1 vs Delta predicted Zelda
Delta binding along PC1 axis
6
4
2
0
−2
−4
−6
−2
−1
0
1
2
3
Delta predicted Zelda (motif)
cor: 0.15
p.value: 0

Figure 7

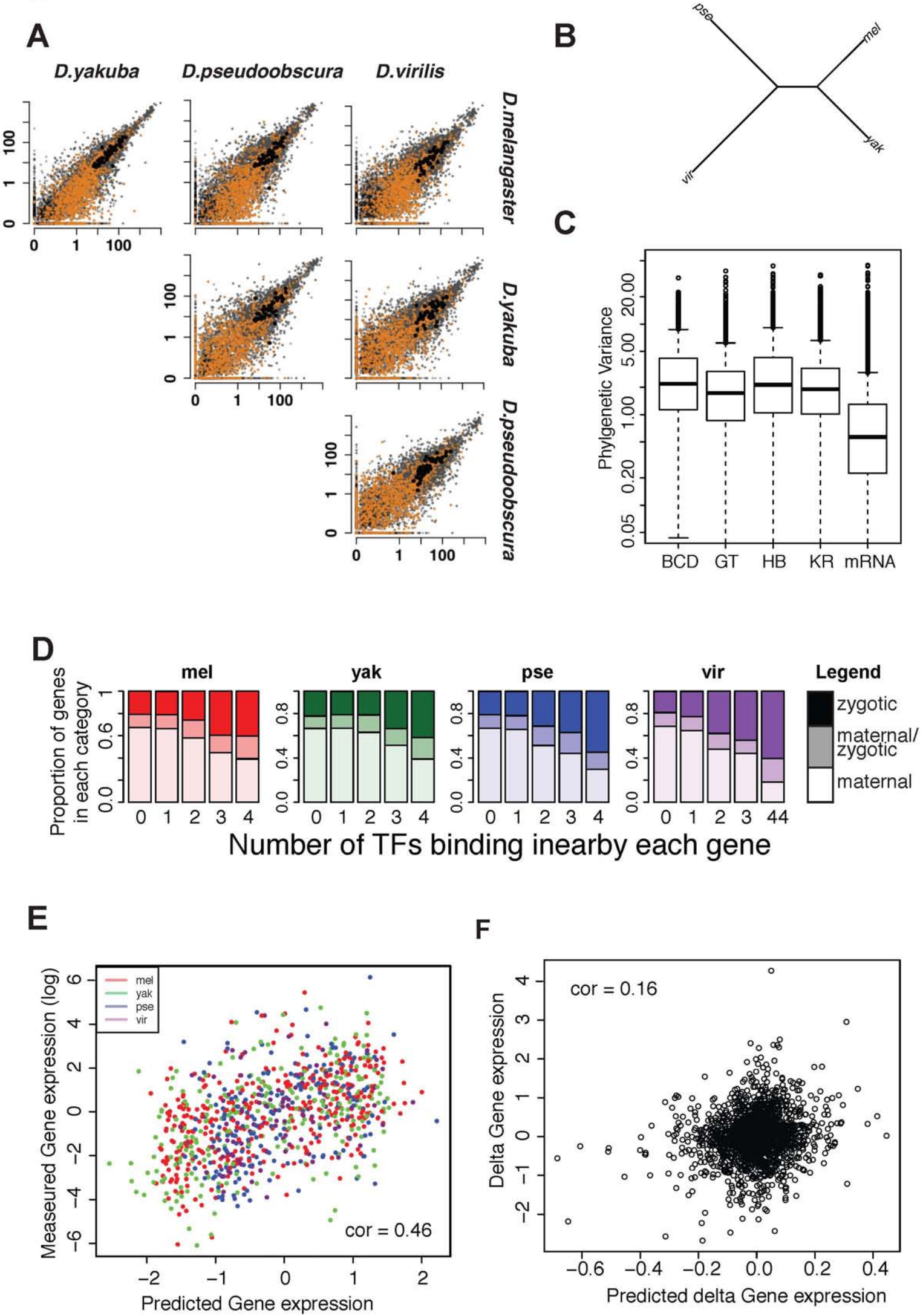

A
D.yakuba
D.pseudoobscura
D.virilis
D.melangaster
D.yakuba
D.pseudoobscura
B
pse
mel
vir
yak
C
Phylgenetic Variance
BCD
GT
HB
KR
mRNA
D
mel
yak
pse
vir
Legend
zygotic
maternal/ zygotic
maternal
Proportion of genes in each category
Number of TFs binding inearby each gene
E
mel
yak
pse
vir
Measured Gene expression (log)
Predicted Gene expression
cor = 0.46
F
cor = 0.16
Delta Gene expression
Predicted delta Gene expression

Figure S1

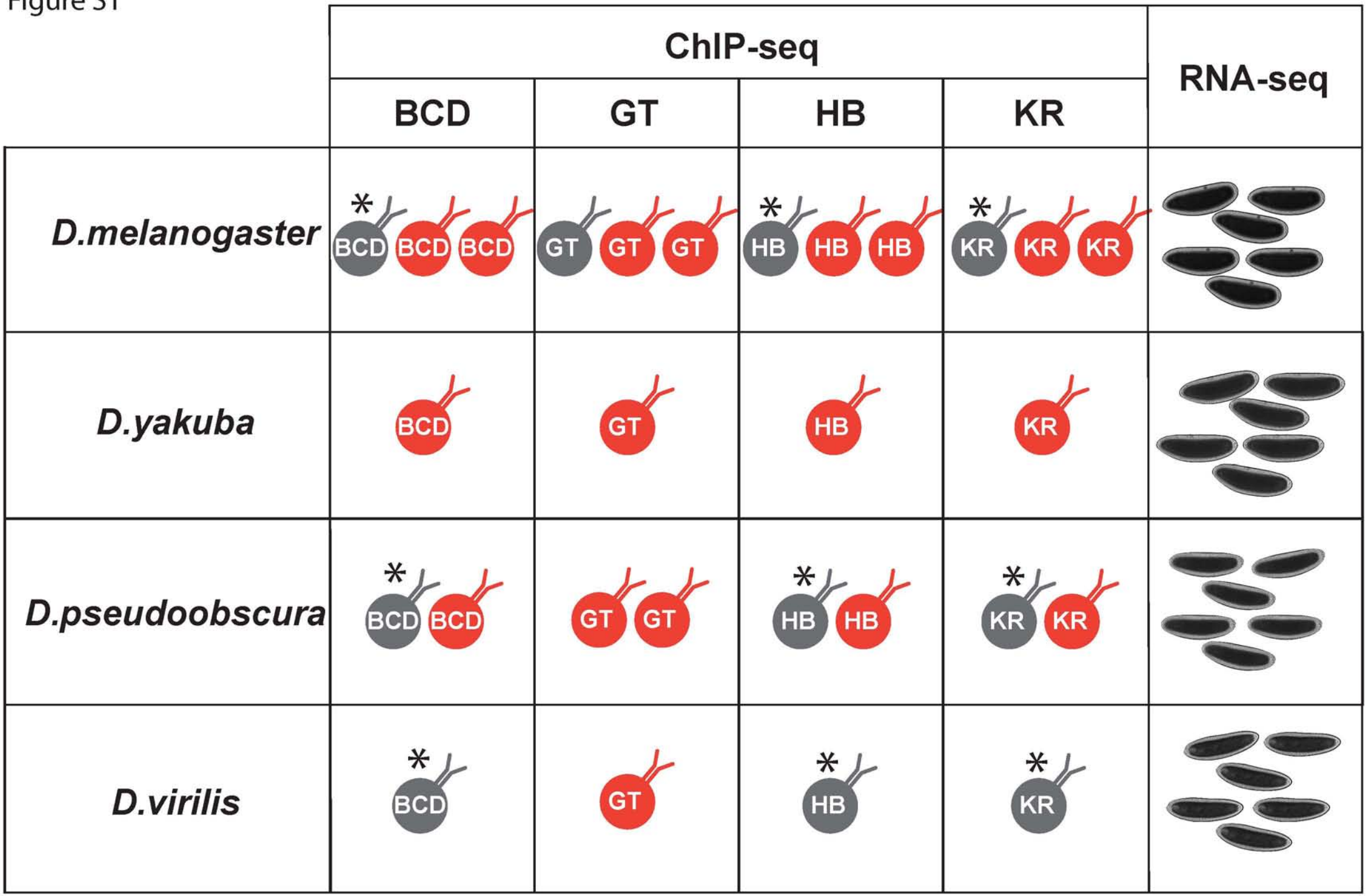
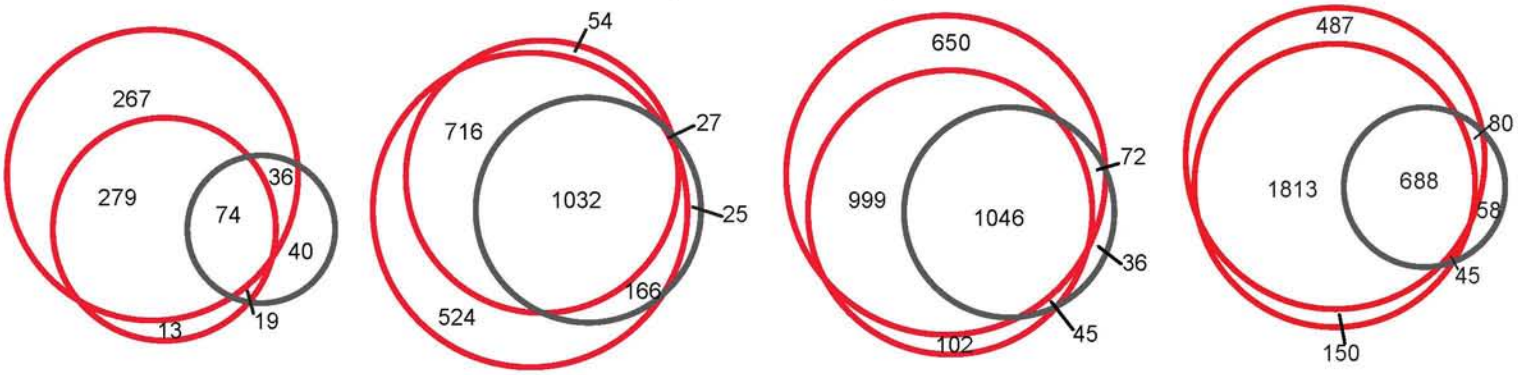

ChIP-seq
RNA-seq
BCD
GT
HB
KR
D.melanogaster
D.yakuba
D.pseudoobscura
D.virilis

Figure S2

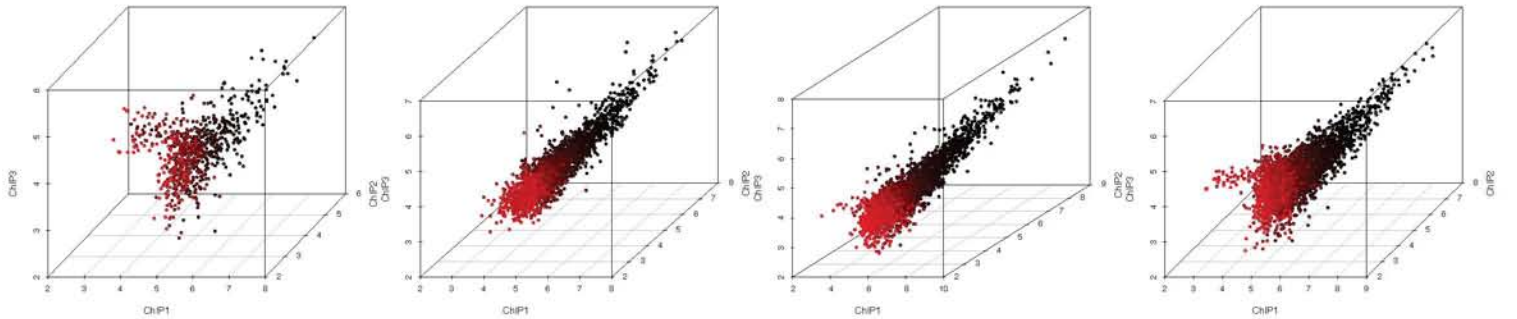
A

BCD GT HB KR

Peak overlaps between replicates



Comparison of occupancy at bound regions between replicates

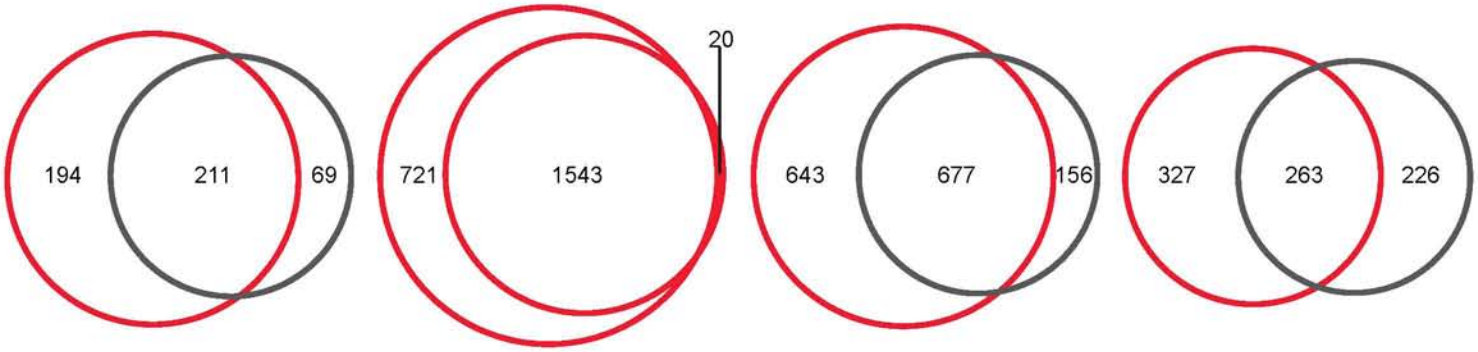


D.melanogaster replicates

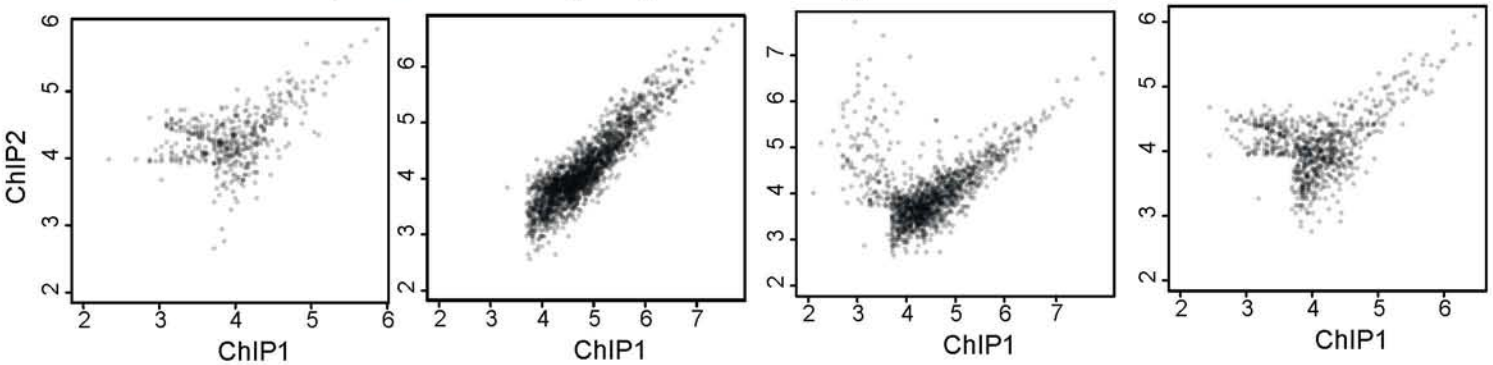
B

BCD GT HB KR

Peak overlaps between replicates



Comparison of occupancy at bound regions between replicates



D.pseudoobscura replicates

Figure S3

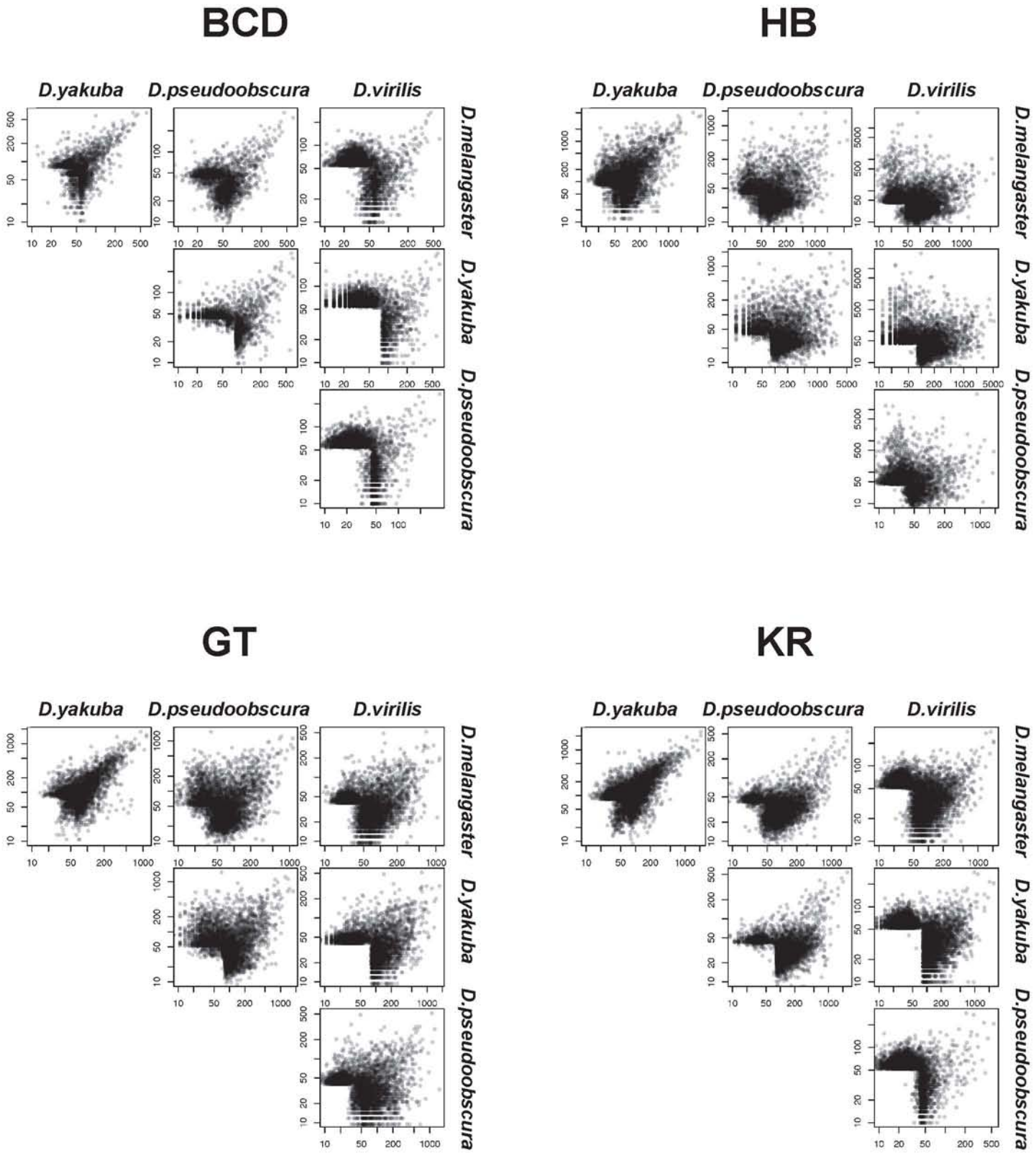

BCD
HB
GT
KR
D.yakuba
D.pseudoobscura
D.virilis
D.melangaster
D.yakuba
D.pseudoobscura

Figure S4

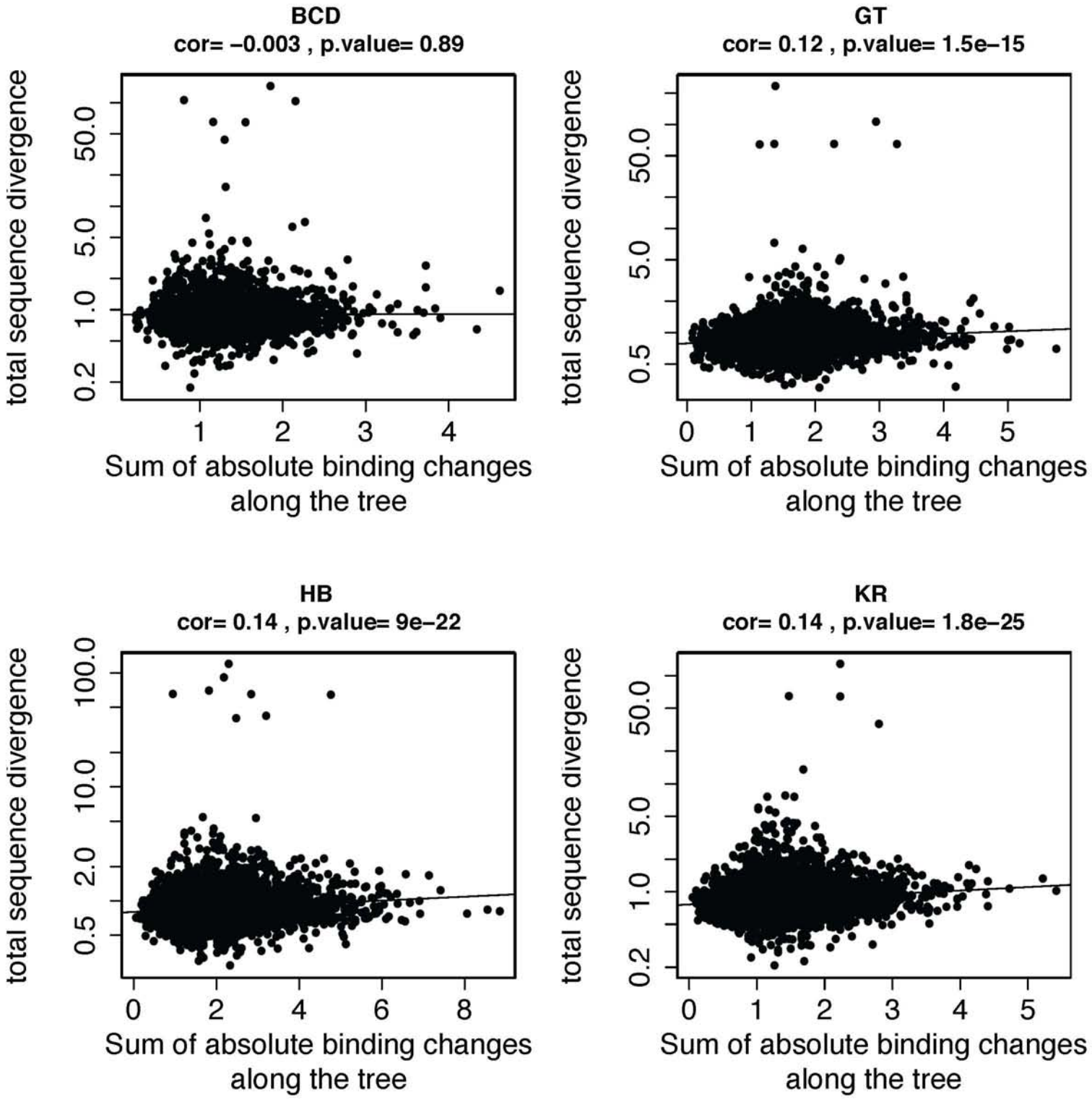

BCD
cor= −0.003 , p.value= 0.89
total sequence divergence
0.2
1.0
5.0
50.0
1
2
3
4
Sum of absolute binding changes along the tree
GT
cor= 0.12 , p.value= 1.5e−15
total sequence divergence
0.5
5.0
50.0
0
1
2
3
4
5
Sum of absolute binding changes along the tree
HB
cor= 0.14 , p.value= 9e−22
total sequence divergence
0.5
2.0
10.0
100.0
0
2
4
6
8
Sum of absolute binding changes along the tree
KR
cor= 0.14 , p.value= 1.8e−25
total sequence divergence
0.2
1.0
5.0
50.0
0
1
2
3
4
5
Sum of absolute binding changes along the tree

Figure S5

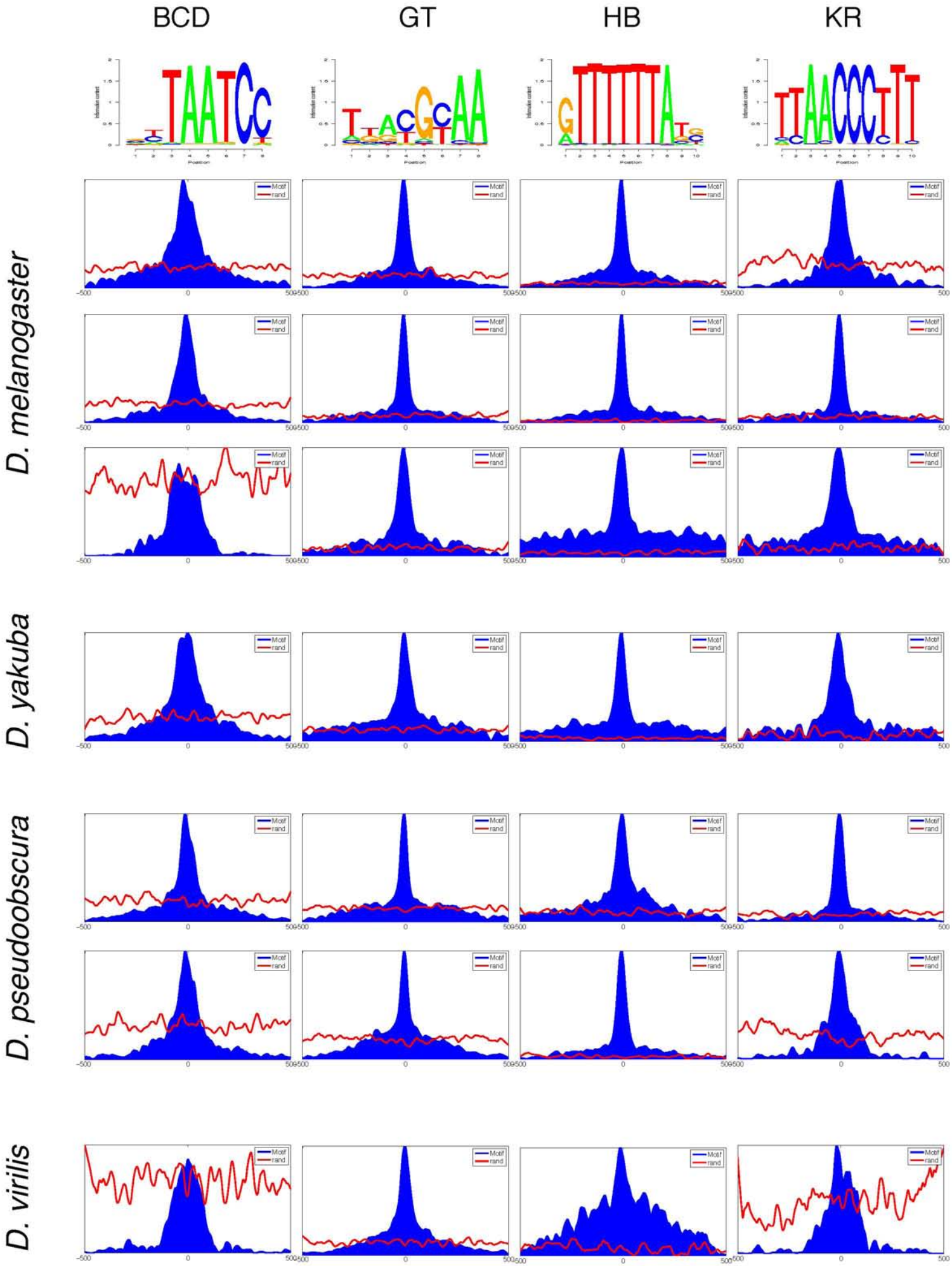

BCD
GT
HB
KR
D. melanogaster
D. yakuba
D. pseudoobscura
D. virilis
Motif
rand
-500
0
500

Figure S6

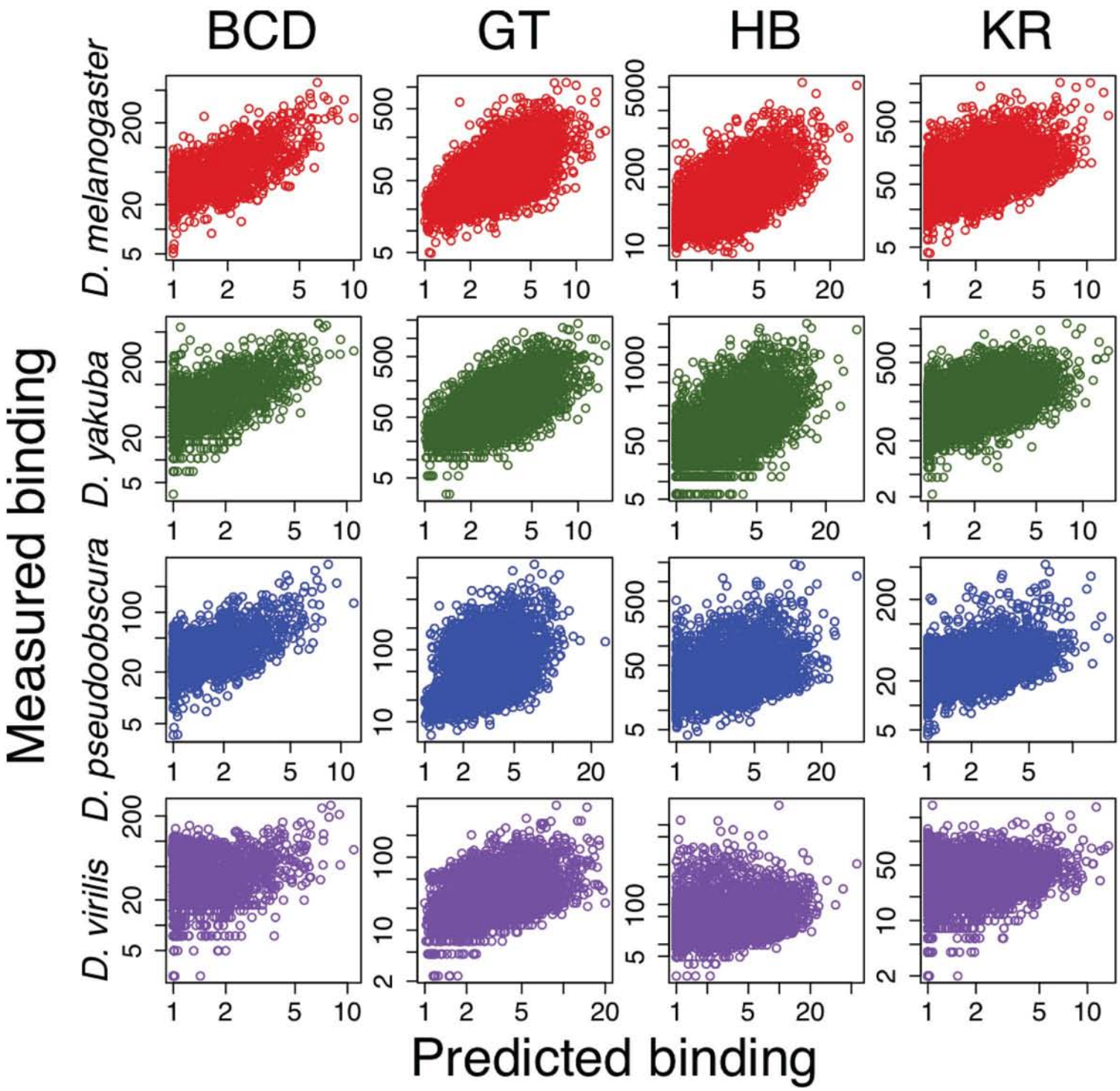

BCD
GT
HB
KR
D. melanogaster
D. yakuba
D. pseudoobscura
D. virilis
Measured binding
Predicted binding

Figure S7

Delta binding (motif) *vs* Delta predicted binding

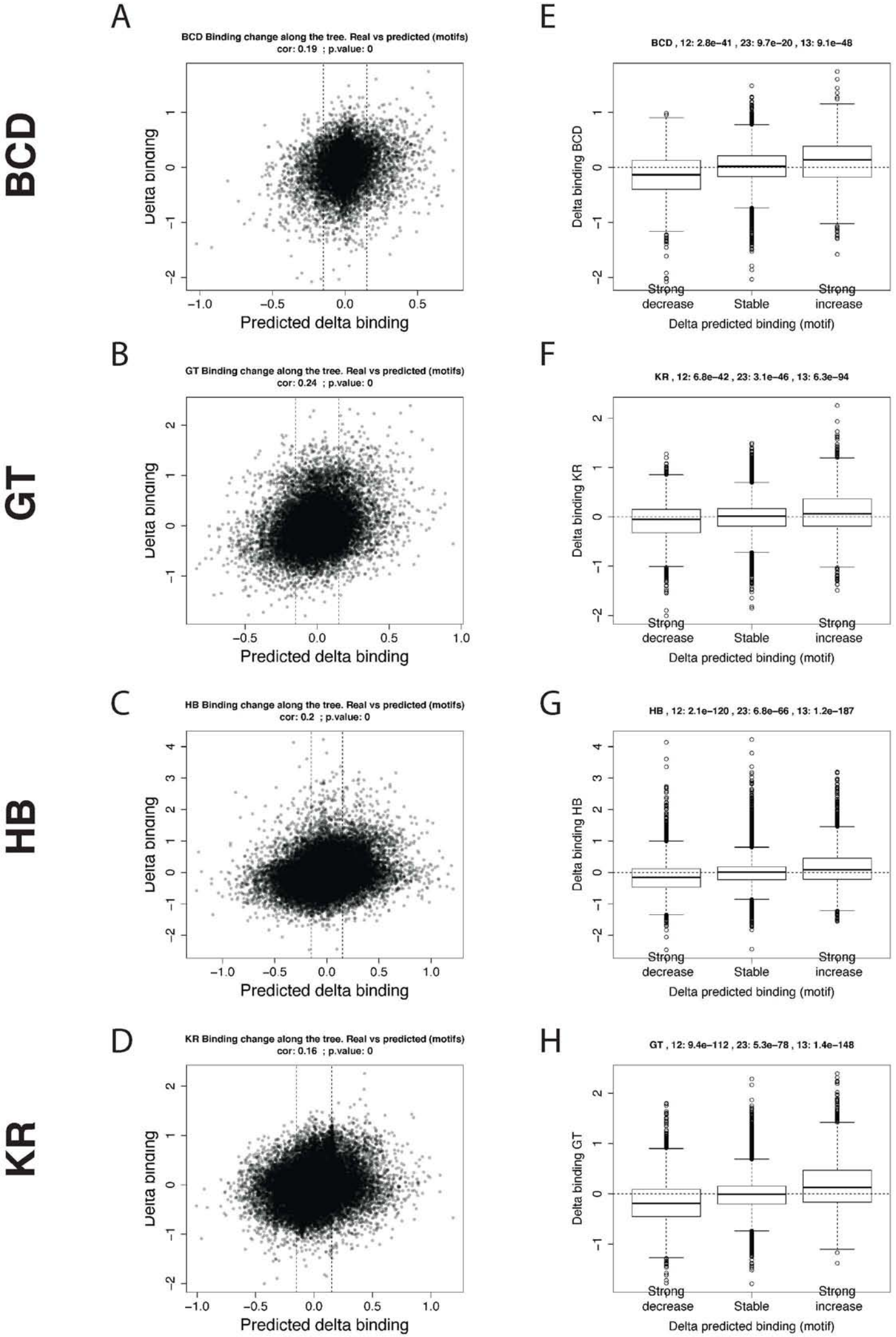


Figure S8

species-specific PCA of TF binding of all factors

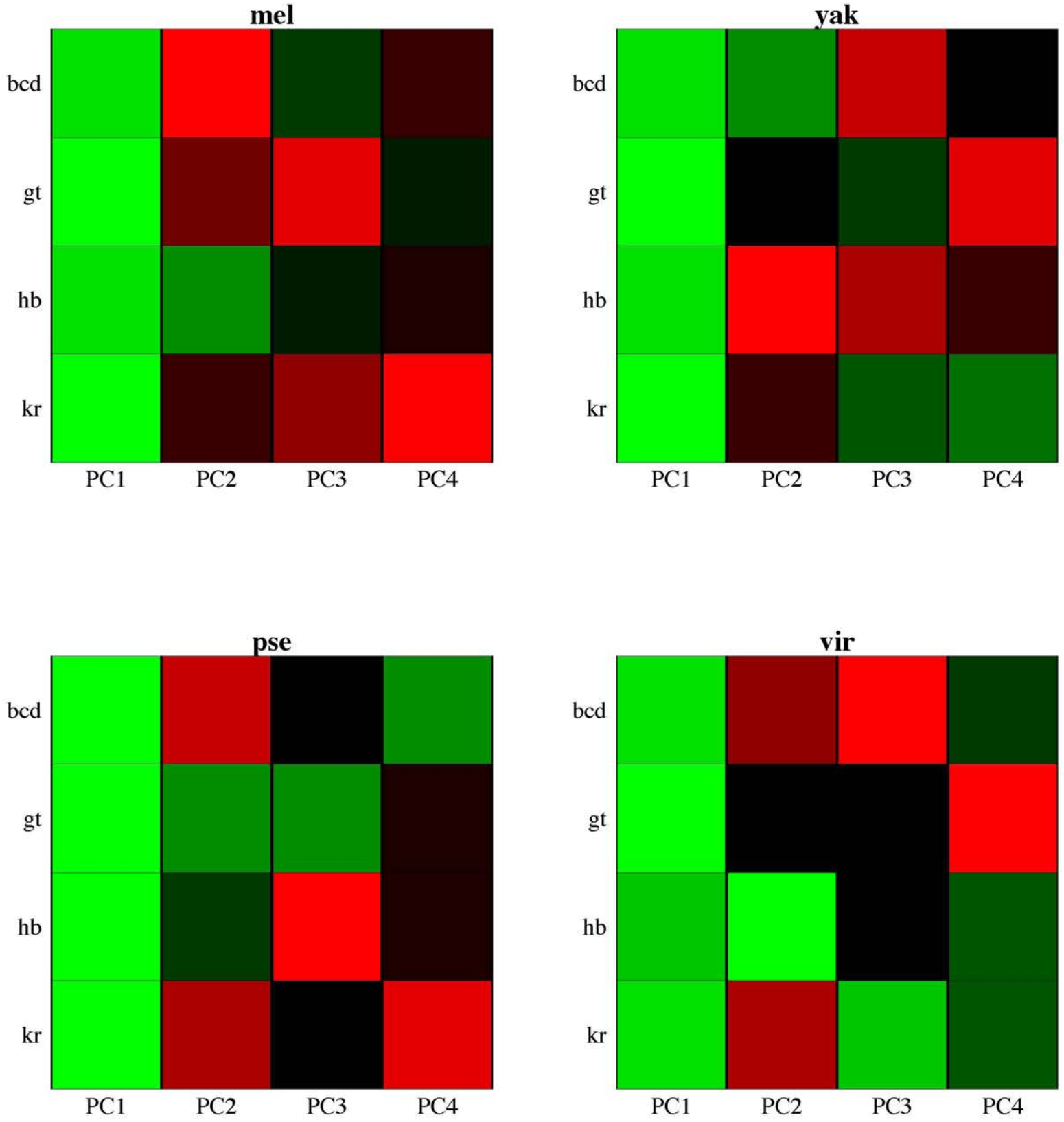
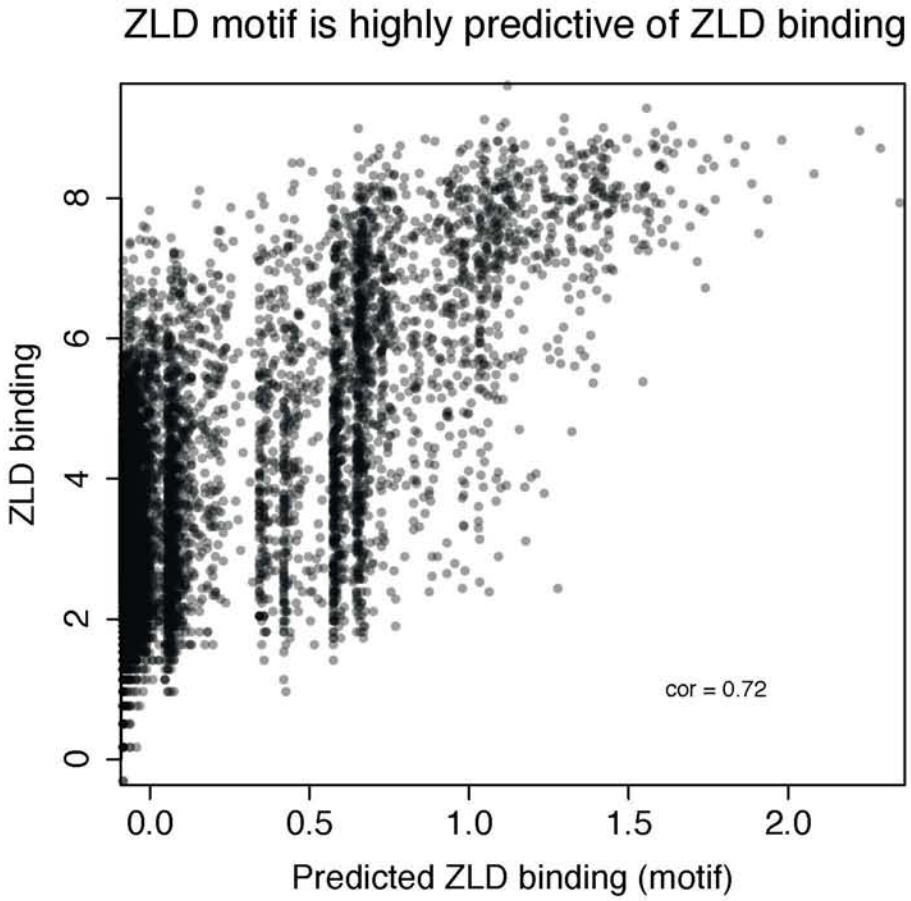


Figure S9

A


ZLD motif is highly predictive of ZLD binding
ZLD binding
0
2
4
6
8
cor = 0.72
0.0
0.5
1.0
1.5
2.0
Predicted ZLD binding (motif)

B

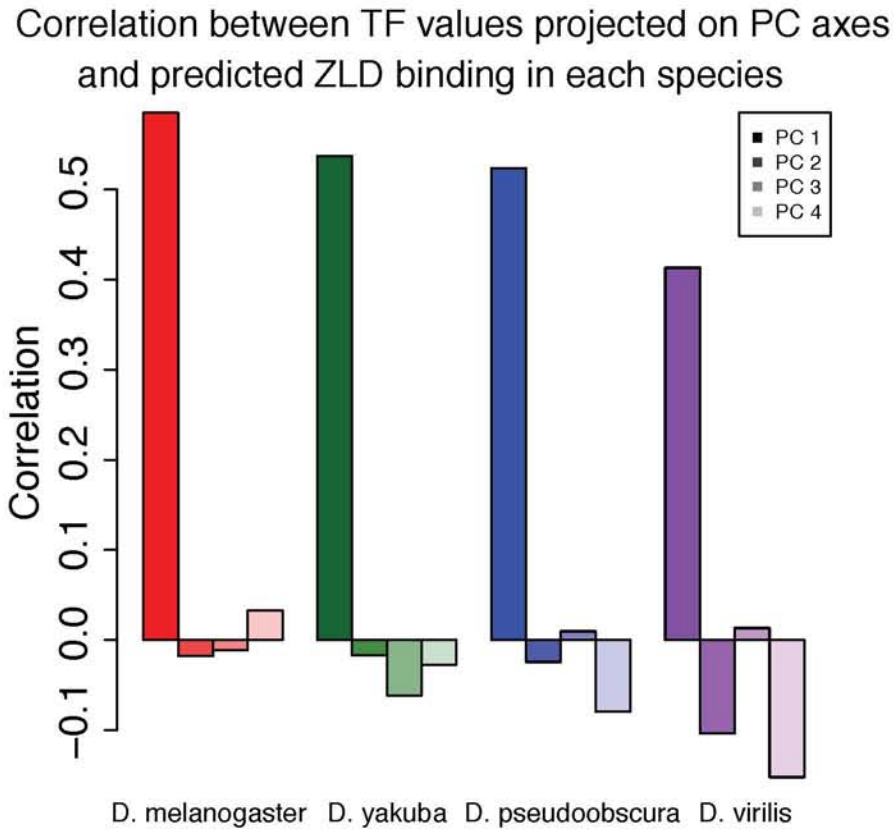

Correlation between TF values projected on PC axes
and predicted ZLD binding in each species
PC 1
PC 2
PC 3
PC 4
Correlation
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
D. melanogaster
D. yakuba
D. pseudoobscura
D. virilis

Figure S10

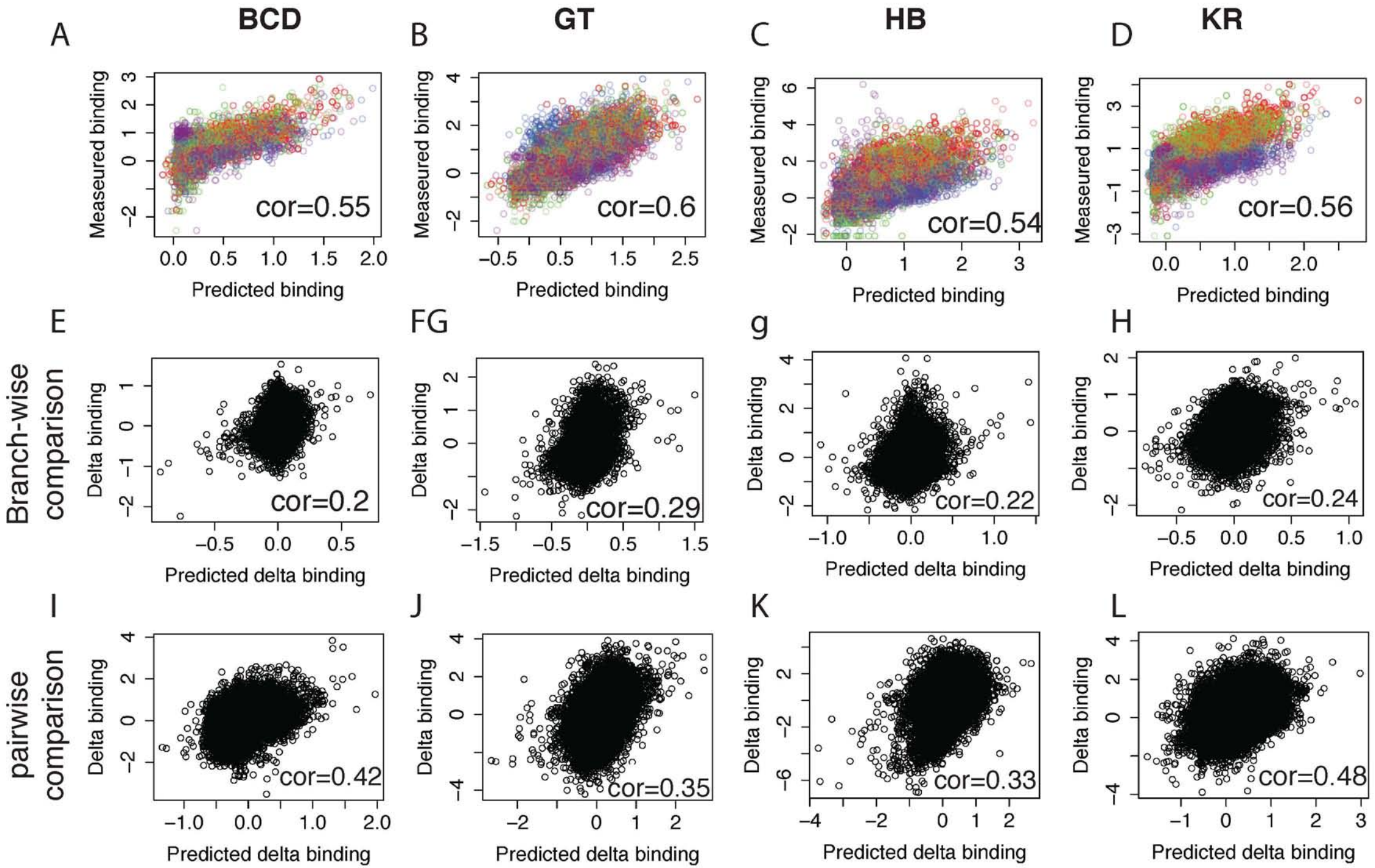

BCD
GT
HB
KR
A
B
C
D
Measeured binding
Predicted binding
cor=0.55
cor=0.6
cor=0.54
cor=0.56
Branch-wise comparison
E
FG
g
H
Delta binding
Predicted delta binding
cor=0.2
cor=0.29
cor=0.22
cor=0.24
pairwise comparison
I
J
K
L
cor=0.42
cor=0.35
cor=0.33
cor=0.48

Figure S11

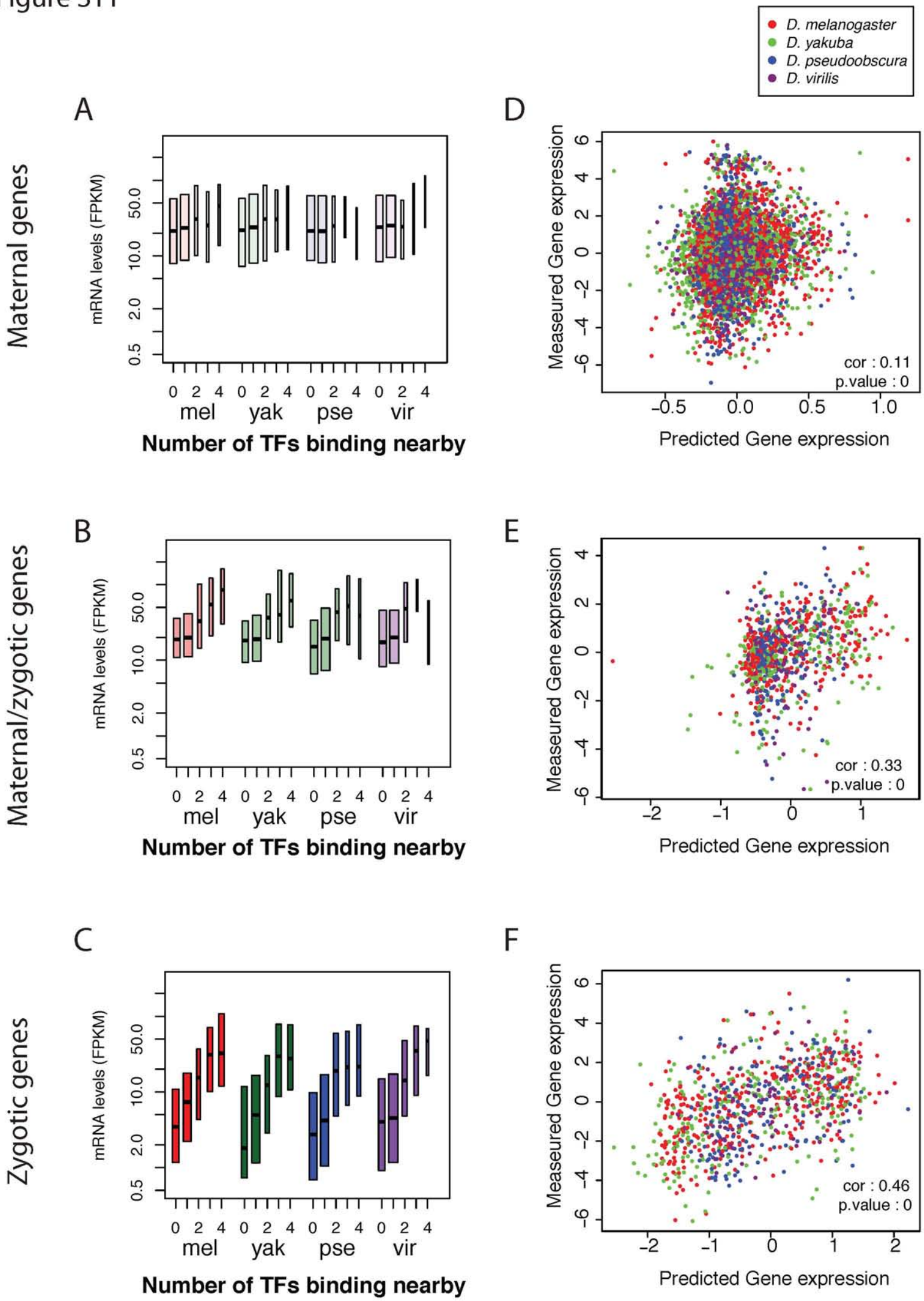

D. melanogaster
D. yakuba
D. pseudoobscura
D. virilis
A
B
C
D
E
F
Maternal genes
Maternal/zygotic genes
Zygotic genes
mRNA levels (FPKM)
0.5
2.0
10.0
50.0
0 2 4 0 2 4 0 2 4 0 2 4
mel yak pse vir
Number of TFs binding nearby
Measeured Gene expression
Predicted Gene expression
cor : 0.11
p.value : 0
cor : 0.33
p.value : 0
cor : 0.46
p.value : 0

Figure S12

Comparison of Delta mRNA levels *vs* predicted Delta mRNA levels
(based on associatedTF binding changes)

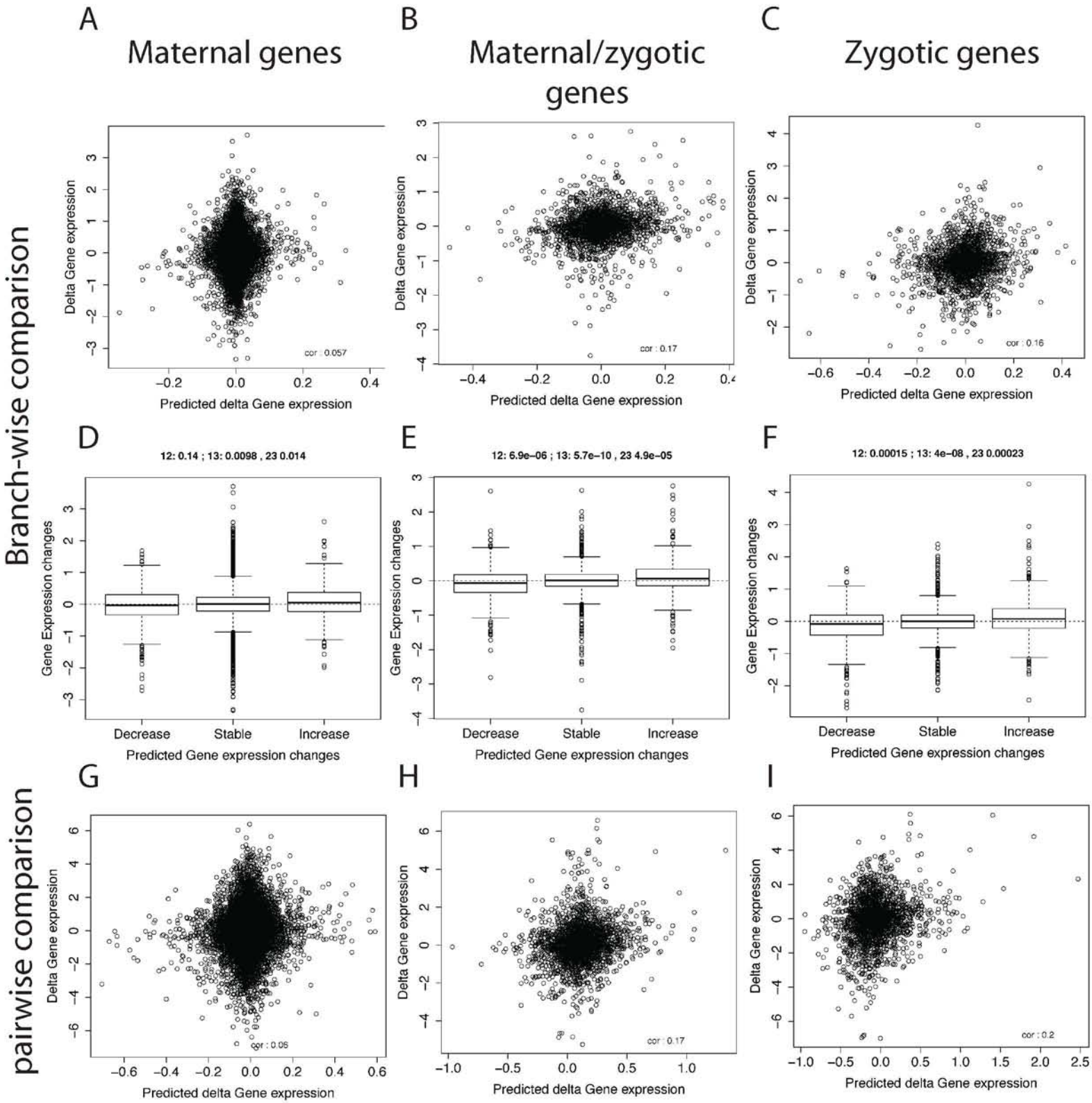
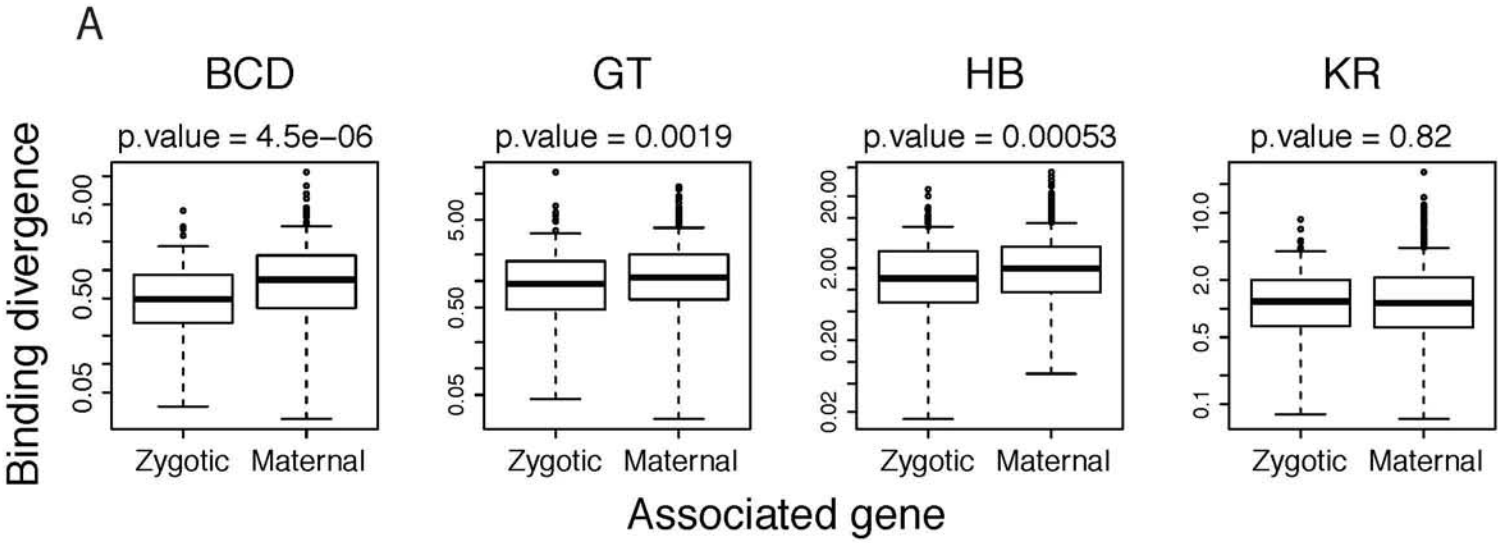
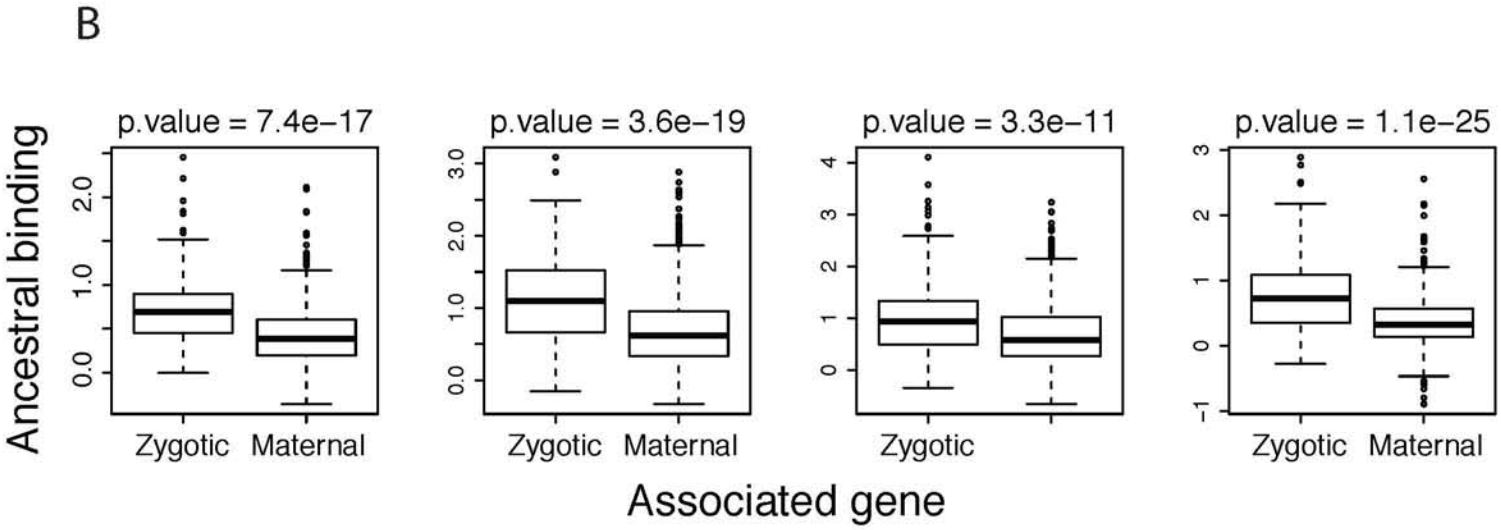


Figure S13


A
BCD
GT
HB
KR
p.value = 4.5e−06
p.value = 0.0019
p.value = 0.00053
p.value = 0.82
Binding divergence
Zygotic
Maternal
Associated gene


B
p.value = 7.4e−17
p.value = 3.6e−19
p.value = 3.3e−11
p.value = 1.1e−25
Ancestral binding
Zygotic
Maternal
Associated gene